



El Niño Agricultural Vulnerability Index in Cordillera for September 2026 - February 2027

(standing crops + number of dry days + historical el niño damage)

Overview & Objective

This map integrates Rice and Corn sector vulnerabilities across the Cordillera Administrative Region (CAR) to guide municipal mitigation, resource pre-positioning, and emergency response during the El Niño dry spell.

Assessment Formula & Methodology

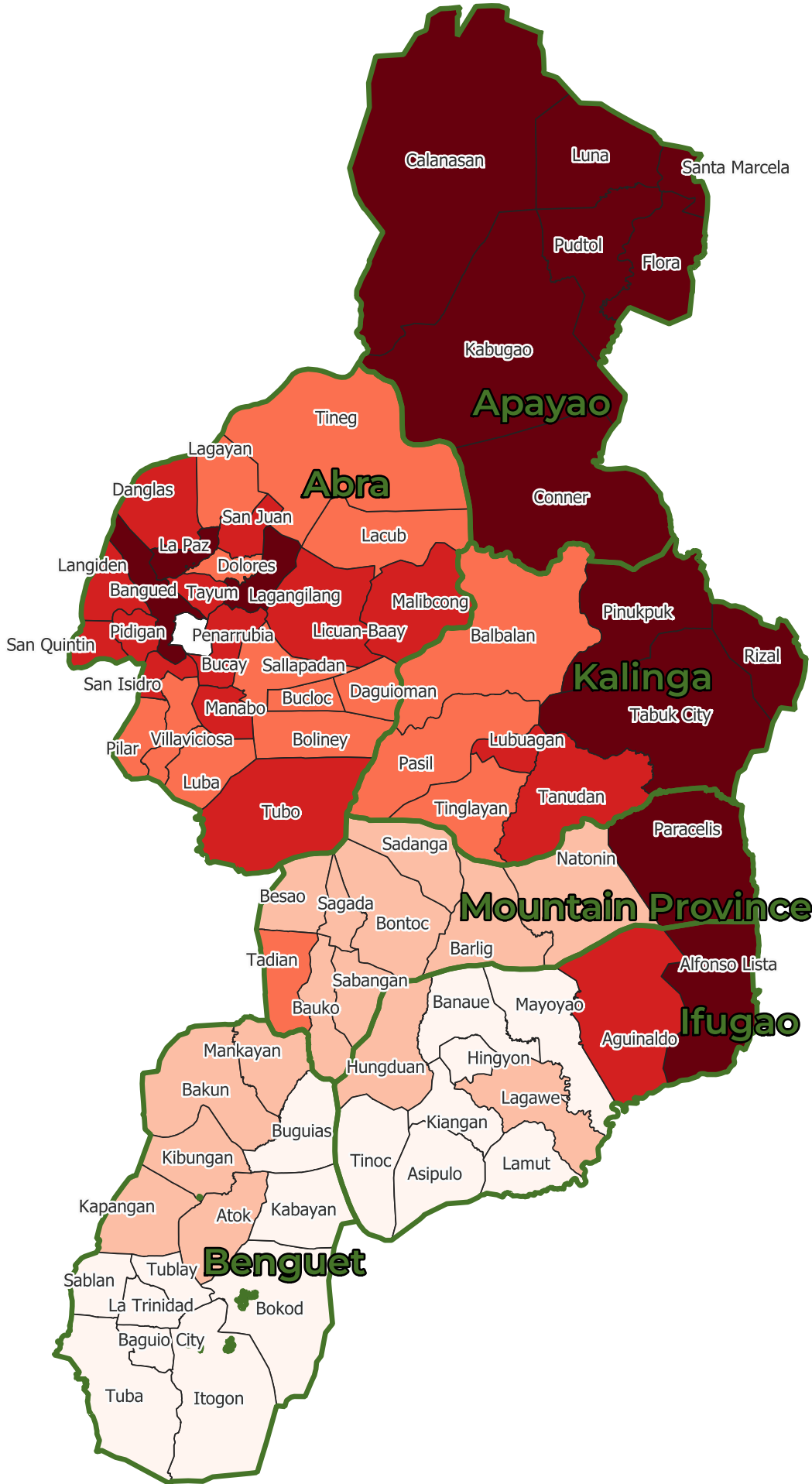
Each municipality's composite index is computed using weighted multi-criteria analysis with Min-Max normalization (0.0 to 1.0).

Criteria & Factor Weights

- Crop Exposure** (50%): Active standing areas for rice (25%) and corn (25%) as of August 15, 2026.
- Historical Sensitivity** (40%): Average yield losses recorded during past El Niño events in 2019, 2024, and 2026.
- Hazard Stress** (10%): Provincial monthly dry-day forecasts for September 2026 - February 2027.

Vulnerability Index Legend

0.000 – 0.044 (Very Low)
0.044– 0.094 (Low)
0.094– 0.143 (Moderate)
0.143– 0.151 (High)
0.151 – 0.227+ (Critical / Hotspot)



Regional Risk Hotspots (Critical Tier)

Based on the combined active standing crop exposure (50%), historical El Niño damage logs from 2019, 2024, and 2026 (40%), and projected provincial dry-day hazards (10%), these areas represent the primary critical risk zones requiring immediate intervention:

- Ifugao (Critical Municipalities: Alfonso Lista, Aguinaldo, and Lamut)**
 - Primary Driver: Massive standing crop exposure—led by extensive corn production zones exceeding 14,900 hectares—combined with high seasonal dry-day hazards and historical drought sensitivity.
- Apayao (Critical Municipalities: Flora, Luna, and Santa Marcela)**
 - Primary Driver: Highest regional dry-day hazard forecast coupled with substantial crop standing stock and severe historical El Niño yield losses.
- Kalinga (Critical Municipalities: Tabuk City, Pinukpuk, and Rizal)**
 - Primary Driver: Extensive active crop production acreage paired with significant historical El Niño damage sensitivity and high seasonal dry-day peaks.
- Abra (Critical Municipalities: Bangued, Manabo, and Pidigan)**
 - Primary Driver: Substantial standing crop exposure across tracking zones combined with documented historical vulnerability and moisture deficits during past dry spells.

Recommended Actions for Hotspot Areas

- Pre-positioning:** Early distribution of drought-tolerant and short-maturity seed varieties in identified high-risk municipal clusters.
- Water Management:** Immediate deployment of small-scale irrigation projects (SSIPs), solar-powered water pumps, and water-harvesting infrastructure in vulnerable production zones.
- Advisory & Support:** Continuous tracking of municipal dry-day hazard metrics via ACAP-CAR to guide timely field advisories and agricultural buffer stock distribution.

Data Sources: DA-CAR Standing Crops Inventory as of August 15, 2026, DA-CAR Corn and Rice Historical El Niño Damage Logs (2019, 2024, 2026), and DOST-PAGASA Climate Projections - Number of dry days as of July 2026.



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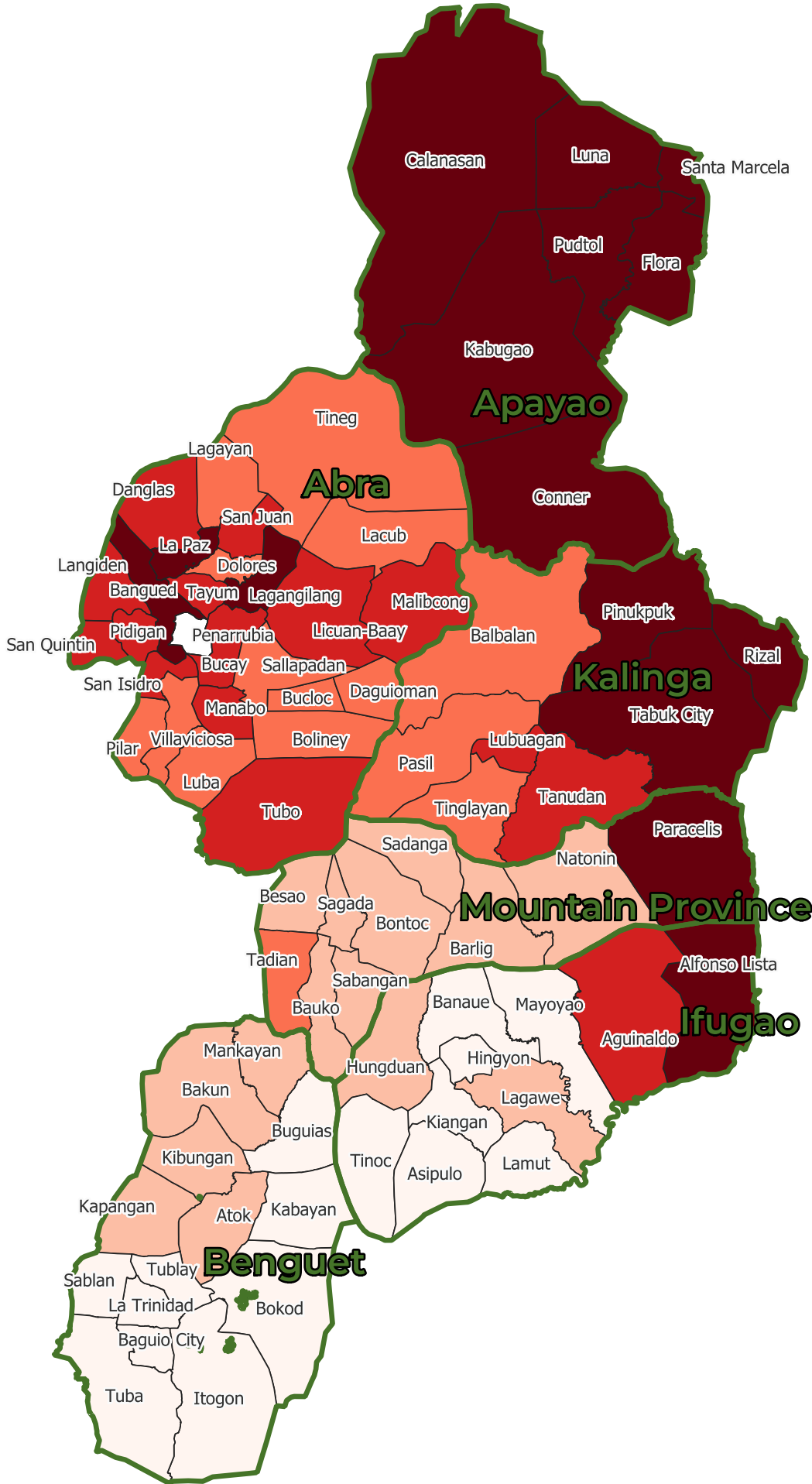
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